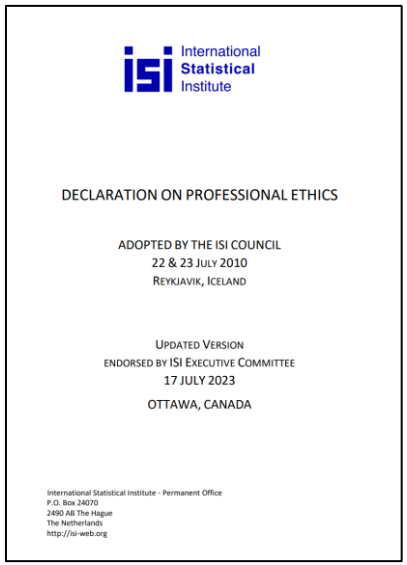
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Analysis of Survival Data

Lecture 7 in-class
Parametric regression
Ethics declaration



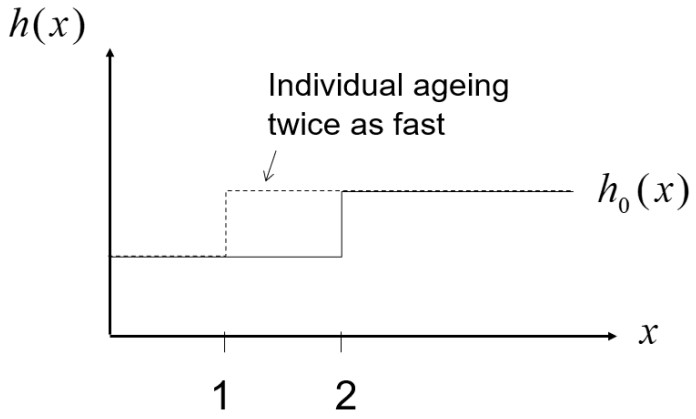


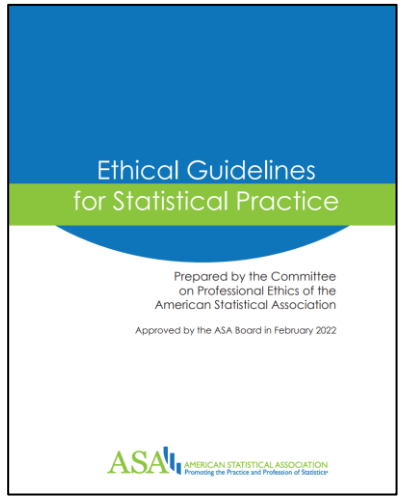
Accelerated failure time model:

$$Y = \mu + \gamma^t \mathbf{Z} + \sigma W$$

$$S(x \mid \mathbf{Z}) = S_0\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right)$$

Acceleration factor (constant for an individual)





Inger Persson

Program L7 in-class

- **Parametric regression**
- **Declaration on professional ethics**
  - Online lecture follow-up
  - Review questions
  - Exercises

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Review questions

What is the main difference between the accelerated failure-time model and the Cox model?

How are the covariates interpreted in the accelerated failure time model?

Review questions

Summarize the pros and cons of the accelerated failure time model and the Cox model

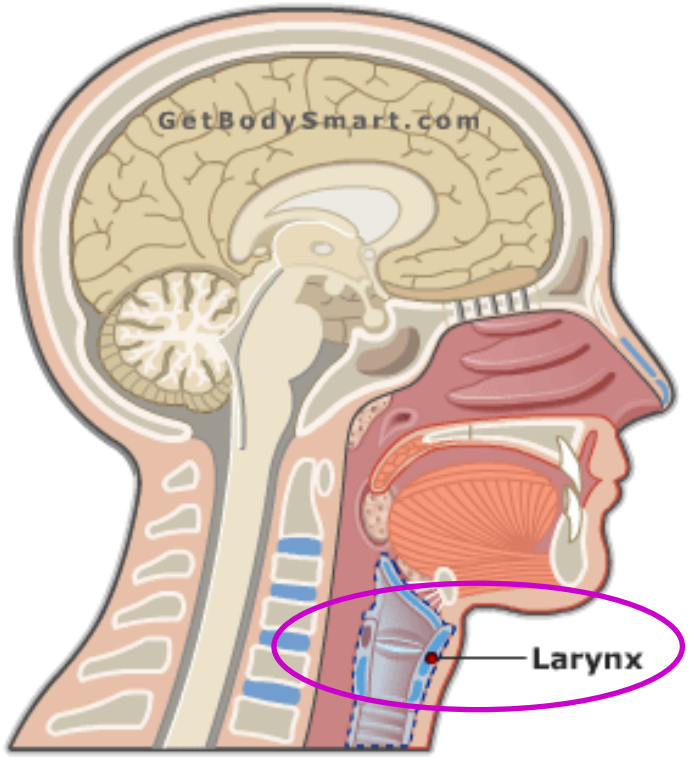
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Review questions

What is the interpretation of the median time to event in the accelerated failure time model?

Example: larynx cancer

A study of 90 males with cancer of the larynx is described in section 1.8.



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Example: larynx cancer

Variables:

X = survival time from first treatment (years)

death = death indicator

stage = Four stages of the disease, stage I – stage IV, ordered from least serious to most serious.

age = patient's age (years)

Example: larynx cancer

1) Fit a distribution to event times only
2) Fit an accelerated failure time model based on the chosen distribution
   - Create dummies for stage
   - Find estimates of γ (for Y = ln X)
   - Transform estimates to β (for X)
3) Let me know when you have interpretable results! I want to hear your interpretations.

Declaration on professional ethics

What do you think is/are the most important part(s) of the ethics documents?

What would you do if you at a future job are asked to come up with a result that supports your boss' opinion (which may not be ethically correct).

Next up

- **Home assignment 2**
  - Feedback will be available before 17/1
- **Oral presentation of home assignm. 2, Task 1**
  - Seminar 18/12
  - See *Oral presentation instructions* in Studium
- **Home assignment 3 (stats students only)**
  - Due 9/1
- **Exam (paper and pen)**
  - 15/1 - sign up in Ladok